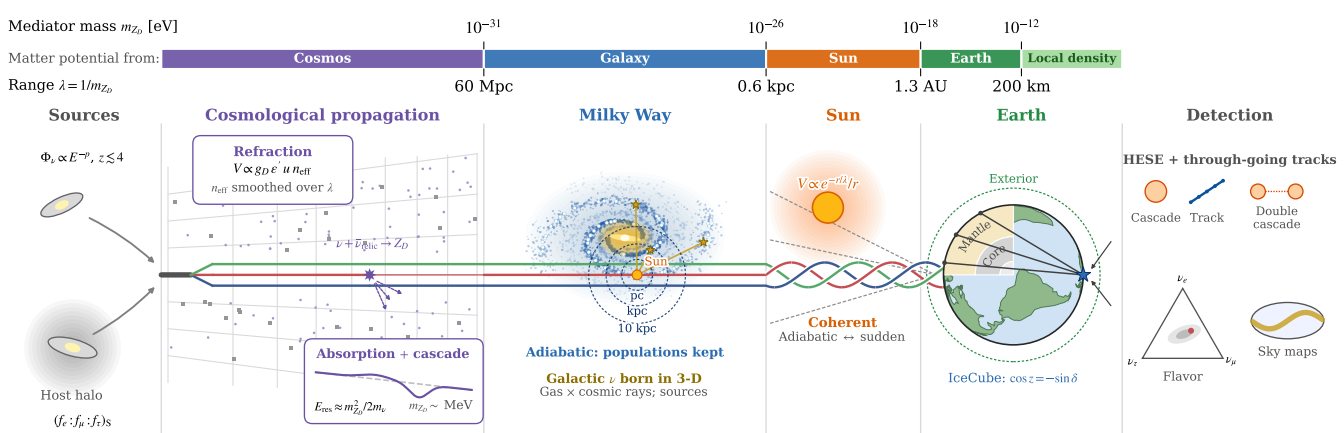

Mediator mass m_{Z_D} [eV]
10^{-31}
10^{-26}
10^{-18}
10^{-12}
Matter potential from:
Cosmos
Galaxy
Sun
Earth
Local density
Range $\lambda = 1/m_{Z_D}$
60 Mpc
0.6 kpc
1.3 AU
200 km
Sources
$\Phi_\nu \propto E^{-\eta}$, $z \lesssim 4$
Host halo
$(f_e : f_\mu : f_\tau)_S$
Cosmological propagation
Refraction
$V \propto g_D \epsilon^{'} u\, n_{\rm eff}$
$n_{\rm eff}$ smoothed over λ
$\nu + \bar{\nu}_{\rm Relic} \to Z_D$
Absorption + cascade
$E_{\rm res} \approx m_{Z_D}^2/2m_e$
$m_{Z_D} \sim$ MeV
Milky Way
Sun
pc
kpc
10 kpc
Adiabatic: populations kept
Galactic ν born in 3-D
Gas × cosmic rays; sources
Sun
$V \propto e^{-r/\lambda}/r$
Coherent
Adiabatic ↔ sudden
Earth
Exterior
Mantle
Core
IceCube: $\cos z = -\sin\delta$
Detection
HESE + through-going tracks
Cascade
Track
Double cascade
ν_e
ν_τ
ν_μ
Flavor
Sky maps
Extragalactic, with z-evolution
Optional host galaxy
Refraction + absorption, one calculation
Thermal relic ν, Debye screening; cascade
$n_{\rm eff}$: Yukawa integral of the Galaxy's neutrons
Galactic ν born in local eigenstates
Potential along paths at all angles to the Sun
Its yearly motion averaged
Yukawa integral of the Earth's neutrons
Exact chords, PREM
IceCube responses
MCEq + Barr
Flavor, spectrum, direction